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Tectonic affinity and evolution of the Alxa Block during the Neoproterozoic: Constraints from zircon U-Pb dating, trace elements, and Hf isotopic composition

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The Jinchuan deposit, which is located in the Longshoushan uplift (LU) on the southwest margin of the Alxa Block (AB), is the third largest Ni-Cu deposit in the world. There are igneous rocks in the LU that widely developed in ~830 Ma, which include ore-bearing ultramafic rock and ore-poor mafic-ultramafic rocks. These rocks were produced in an intracontinental extension-rift environment, which is likely related to the Rodinia supercontinent (RS) breakup. However, contemporary magmatic rocks are absent in the AB, except the ~830 Ma igneous rocks in the LU. In this study, we present two types of metamorphic mafic-ultramafic rocks at the border of Jinchuan II and IV mineral areas, which is inside the F₁₇ fault zone. One type is a red-coloured altered rock formed by ultramafic rock oxidation; there are a small number of zircons with multiple ages in the rock, which are considered to be inherited and metamorphic (hydrothermal) zircons. The other type is a chlorite tremolite schist formed by the metamorphism of fault clastics; there are a large number of zircons developed in this rock, which are mostly magmatic zircons with ages between 900 and 800 Ma and a few hydrothermal zircons. Analysis of the trace elements, the 900–800 Ma zircons, have high (Sm/La)_N, low Ce/Ce*, and high Th/U values, which fit the characteristics of typical magmatic zircons. There are also high Y, U, HREE, Nb, and Ta values, which indicates that the host rocks were mostly larvikite and dolerite, and high Y, Ce, Yb, Th, Gd, and Er values indicate a within-plate basaltic origin. All these 900–800 Ma zircons have a $\epsilon_{\text{Hf}}(t)$ in the range of –2 to –7, which is similar to the Jinchuan ore-bearing ultramafic rock with a partial melting origin of enriched mantle. The absence of detrital zircon age peaks in the Neoproterozoic indicates that the AB has been separated from the North China Craton since the Neoproterozoic to Paleozoic. The detrital zircons in the Yangtze Block (YB) strata with Neoproterozoic ages are slightly younger than those in the AB, with more depleted $\epsilon_{\text{Hf}}(t)$ values, which is similar to the zircons from the Jiangnan Orogen igneous rocks. All this evidence indicates that the AB may have experienced at least 100 Myr of intracontinental extension-rift environment from 900 to 800 Ma during the Neoproterozoic and participated in the breakup of the RS. However, the location of the AB during the Rodinia may not have been beside the YB.

KEYWORDS

detrital zircon, Hf isotope, Jinchuan, the Alxa Block, zircon U-Pb dating

1 | INTRODUCTION

The Jinchuan Cu-Ni sulphide deposit is located in the Longshoushan uplift (LU) along the southwest margin of the Alxa Block (AB), with reserves of 0.5 billion ton of sulfide ore and is considered to be the third largest Cu-Ni sulphide deposit in the world. The Jinchuan deposit is a typical ultramafic magmatic liquation deposit with a high mineralization rate, in which 47.8% of the ultramafic intrusions are mineralized. Since its discovery in 1958, many researchers have studied the Jinchuan deposit. Li, Su, Song, and Liu (2004); Tian et al. (2007); Zhang, Kamo, Li, Hu, and Ripley (2010); and Jiao et al. (2014) used zircon U-Pb dating to constrain the age of the Jinchuan ore-bearing ultramafic rock, obtaining 827, 807, 831, and 840 Ma, respectively, which indicates a Neoproterozoic formation age. Yang, Du, Lu, Qu, and Chen (2005); Yang, Chen, Qu, Yang, and Du (2007); and Keays, Ihlenfeld, McInnes, Zhou, and Lambert (2004) used sulphide ore Re-Os dating to constrain the mineralization age, obtaining 833, 840, and 897 Ma, respectively, which indicates the sulphide ore formed at the same time as the ultramafic rock and may have undergone some crustal contamination.

Tang and Bai (1999) suggested that during the Mesoproterozoic, the AB was on the southwest side of the North China Craton (NCC), which was an extensional environment of a passive continental margin. Zeng (2014) researched the high-field-strength elements of the Jinchuan ore-bearing ultramafic rock and suggested that the Jinchuan ultramafic rock was formed during an initial period of intracontinental extension-rift in the Neoproterozoic and may be related to the breakup of the Rodinia supercontinent (RS). Li et al. (2005) compared the Jinchuan ultramafic rock, which has a mafic dike cutting through it, with basic rocks related to the mantle plume along the north margin of the Yangtze Craton, and suggested that the Jinchuan ore-bearing ultramafic rock is related to the mantle plume activity that accompanied the RS breakup and that the LU belonged to the Yangtze Craton rather than the NCC prior to the RS breakup. However, Duan (2015) compared the basement stratum of the LU with the NCC and suggested the LU was part of the NCC since the Mesoproterozoic, whereas the Jinchuan ultramafic rock originated from the high Mg basaltic magma of the NCC.

The exposure of a mass of Meso-Neoproterozoic mafic-ultramafic intrusions in the Jinchuan mineral area and the AB constitutes the Longshoushan mafic-ultramafic dike swarm. Previous research found that these mafic-ultramafic rocks have the same formation age of ~830 Ma as the Jinchuan ore-bearing ultramafic rock, and ages from other time periods in Neoproterozoic have been rarely reported, which makes it difficult to constrain the tectonic evolution of the AB during the Neoproterozoic. However, some researchers compared the basement strata of the LU and NCC, suggesting that the AB was amalgamated with the NCC during the Mesoproterozoic (Duan, 2015). Nevertheless, the Longshoushan area experienced wide development of basaltic magmatic events during the Neoproterozoic, while

few magmatic events have been reported in the NCC. Some researchers suggested that the AB was an independent block, which separated from the RS during the Neoproterozoic and amalgamated with the NCC during the Ordovician (Zhang, Li, Liu, & Feng, 2011). Other researchers compared the age peaks of the detrital zircons in the Paleozoic strata of the AB and NCC, and it was suggested that the amalgamation between the AB and NCC occurred during the late Devonian to early Carboniferous (Yuan & Yang, 2015; Zhang, Zhang, Zhang, et al., 2016; Zhang, Zhang, & Zhao, 2016).

Despite these viewpoints, it is not controversial that the AB was formed in an intracontinental extension-rift environment during the Neoproterozoic. However, there are still no age constraints on the evolution of the extension-rift environment. In this study, we discovered two kinds of metamorphic mafic-ultramafic rocks from Jinchuan mineral area, analysed the zircon U-Pb-Hf isotopes and trace elements, and found a multiperiod of magmatic events during the Neoproterozoic. We have also suggest that the AB experienced a continental breakup and constrain the breakup age. We further compared the ages and Hf isotopic composition of the detrital zircons from multiple blocks and suggest the AB location in the RS was not beside the Yangtze Block (YB). All these discoveries may provide a new understanding of tectonic affinity and evolution of the AB during the Neoproterozoic.

2 | GEOLOGICAL BACKGROUND

The AB is shaped like an inverted triangle and located in the western part of the NCC (Figure 1a). The northern margin of the AB is the Paleozoic Central Asia Orogenic Belt, and the southern margin is the ~450 Ma North Qilian Orogenic Belt (Song, Niu, Su, & Xia, 2013). Zircon age peaks and strata comparisons indicate that the AB and eastern Ordos Block were amalgamated along the ~1.95 Ga Khondalite Series during the Paleoproterozoic, which is considered to be a Paleoproterozoic collision orogenic belt (Hu, Gong, Wu, Liu, & Liu, 2014; Tang, Li, Zhang, Edward, & Wang, 2014; Zhao, Sun, Simon, & Li, 2005). The AB has a typical dual structure of metamorphic basement and sedimentary cover; the basement is composed of 2.7~1.9 Ga amphibolites and granitic gneiss (Gong et al., 2016); and the sedimentary covers are strata from the Cambrian to middle Ordovician.

The Longshoushan area is located on the southwest margin of the AB, the main part of which is distributed from NW to SE as a narrow-banded shape, and it gradually turns to an E-W direction near the Jinchuan mineral area (Figure 1b). This district is separated from the Tarim Block by the Altun Tagh fault zone in the west, separated from the Khondalite Series by the Bayanwula fault zone in the east, and separated from the Hexi corridor transitional zone and Chaoshui Basin by the south and north Longshoushan faults (Zeng, 2016). The strata of the Longshoushan area are composed of the Longshoushan group

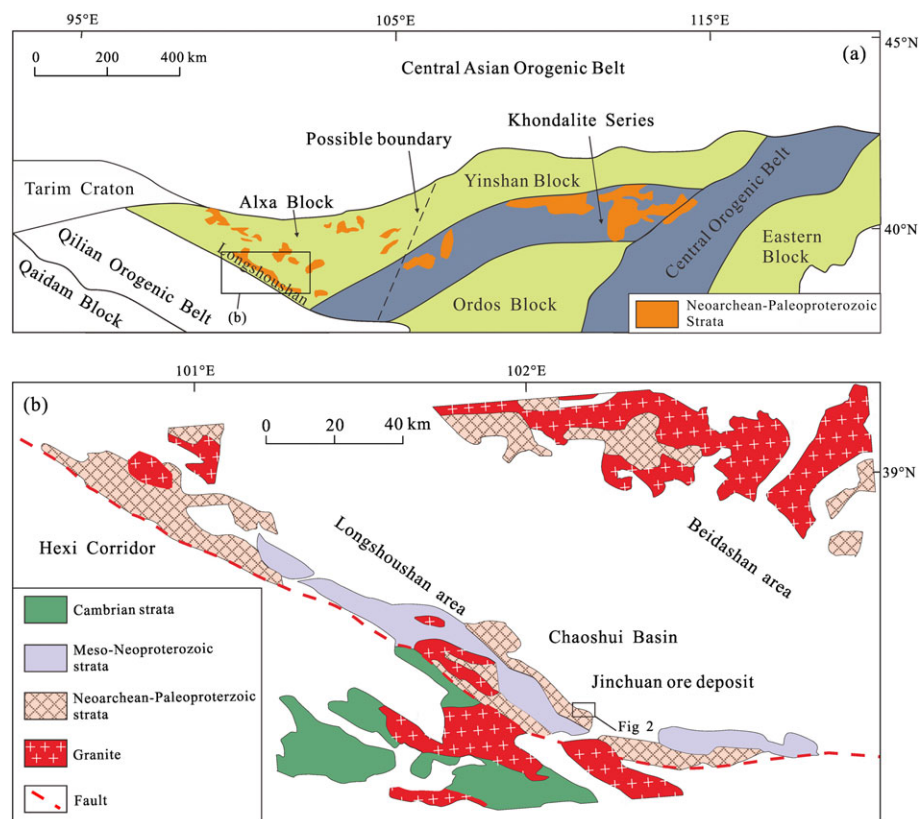


FIGURE 1 Regional geological map of the (a) Alxa Block and (b) Longshoushan area (after Zeng, 2016) [Colour figure can be viewed at wileyonlinelibrary.com]

and overlying cover. The Longshoushan group is the oldest basement in the district which is composed of migmatitic granite, marble, gneiss, and amphibolite. Meso-Neoproterozoic strata are occasionally exposed, which is composed of metamorphic sandstone and limestone, with multicycle. In addition, the cover layers of the district are strata from the Devonian, Carboniferous, Permian, Jurassic, Cretaceous, Neogene, and Quaternary (Bureau of Geology and Mineral Resources of Gansu Province [BGMGRP], 1981).

Magmatic events are abundant in the Longshoushan area, granites are mostly developed, and the existing reports of zircon U-Pb ages can be divided into LU granites from two periods: The Paleoproterozoic granites distributed in pre-Statherian strata, which have a single lithology of metasomatic granite with ages of 2.2–2.0 Ga (Xiu, Lu, Yu, & Yang, 2002; Xiu, Yu, Li, Zuo, Li, & Cao, 2004). The Paleozoic granites are widely distributed over a large area with long and multiple age periods and multiple lithologies; they are mainly composed of monzonitic granite and diorite granite with ages of 570–401 Ma, which is related to the tectonic subduction events and the collision and post-collision of the North Qilian Block (Nie et al., 2016; Wei et al., 2013; Zhao, Zhang, Tang, Yao, & Yang, 2016). There are approximately 20 mafic-ultramafic rock masses discontinuously distributed in the Longshoushan area, which comprise an important mafic-ultramafic rock belt. The Jinchuan ore-bearing mafic-ultramafic intrusion is the best-known in the district; the intrusion is distributed in the Precambrian Baijiazui Formation with a NW-SE direction and multiple lithologies from dunite to pyroxenite (Tang & Li, 1996). The age of the Jinchuan intrusion was long debated, but in recent years, many studies using high-precision zircon U-Pb dating and rich sulphide ore Re-Os

dating indicate that the Jinchuan deposit has a diagenesis-mineralization age of between 850 and 820 Ma. Duan (2015) researched the Xijing mafic intrusion northwest of Jinchuan mineral area, suggesting that the intrusion is mainly composed of olivine, pyroxenolite, and gabbro, which formed at 421 Ma as a result of the continuous subduction of the Qilian Ocean, upwelling of asthenosphere, and crustal contamination. The Maocaoquan mafic-ultramafic intrusion is exposed along a 3 km distance from the southwest of the Jinchuan ore-bearing ultramafic rock. Jiao et al. (2012) used the zircon U-Pb dating to obtain an emplacement age of 832.5 Ma and combined with whole-rock geochemistry, Sr-Nd-Pb isotopes; it is considered to be of co-magmatic origin with the Jinchuan intrusion. The Xiaokouzi intrusion is composed of serpentinite and altered peridotite. Jiao et al. (2017) used zircon U-Pb dating to obtain a formation age of 284.4 Ma, which was an extension environment after the closure of the paleo-Asian Ocean. Duan et al. (2015) studied the widespread dolerite dike in the Jinchuan mineral area and suggested that these dikes cut through the Jinchuan ore-bearing intrusion about 424 Ma, which resulted in a post-collision environment after the closure of the Qilian Ocean and a high degree of crustal contamination.

3 | FIELD GEOLOGY AND SAMPLE DESCRIPTION

The Jinchuan Cu-Ni sulphide deposit is located in the LU, along the southwest margin of the AB, and the ore-bearing ultramafic rock intrudes into the pre-Statherian Baijiazui Formation as a dike. The

Baijiazui Formation is mainly composed of migmatite, gneiss, and marble, and the marble and migmatite are the main wall rocks of the ore-bearing intrusion. Igneous rocks are widely developed in the area, from acidite rock to ultrabasic rock, and plutonite rock to subvolcanic rock, which are all exposed. Ultrabasic rock is mostly developed among these rocks and exposed as rock masses, lentils, and dikes. Acidite rocks are secondary, which are developed and exposed as stocks and dikes. Intermediate-basic rocks are less developed, with a small scale and usually exposed as dikes.

The Jinchuan ore-bearing ultramafic intrusion is exposed as a dike, which has a length of approximately 6,500 m, width of 20–500 m, NW trend, and 73° dip angle; the whole dike extends for several kilometres. The later NE-trending fault cuts the intrusion in several sections, and according to the discovered exploration sequence, the four mineral areas are divided into III, I, II, and IV from west to east. The main lithology of the Jinchuan ultramafic rock is lherzolite, followed by pyroxene peridotite in the middle part of the rock mass, plagioclase lherzolite, olivine websterite, and olivine pyroxenolite on the edge of the rock mass; these five lithologies have no obvious boundaries. The Jinchuan intrusion has a characteristic olivine cumulates, and the highest volume fraction can reach 80% in conglomerate. The alteration of the Jinchuan intrusion is mainly serpentinization, followed by tremolitization, actinolitization, chloritization, steatitization, and carbonatation. On the binding site of the ore bodies, tectonic activities are intense, and the rock is almost completely altered.

The F_{17} fault is a transtensional fault with an NE trend, a length of 1,000 m along the trend, 130° tendency, and a nearly 73° dip angle. The fault is composed of a fracture zone, which has a width of 2–6 m. The fillings are related to the cutting strata. From SW to NE, the F_{17} fault cuts through the pre-Statherian Baijiazui Formation of banded-homogeneous migmatite, ore-bearing ultramafic rock, and Baijiazui Formation serpentine marble (Figure 2).

Near the ultramafic rock cutting through the F_{17} fault, dolerite dikes are distributed along the fault, with a rich abundance of ore. We discovered two types of metamorphic rocks: One type is a red-coloured altered rock, which is in contact with the ultramafic rock

and has a gradually changed colour from dark green to red brown. The red-coloured altered rock is hard and compact and usually cross cut by later quartz-calcite veins. The other type of rock is a greenschist facies, which is grey to white in colour, compact and broken, and schistosity is obvious. This rock may have metamorphosed from the fault clastics. These two types of rocks are only exposed near the junction of the F_{17} fault, ore-bearing ultramafic rock, and the Baijiazui Formation (Figure 3a,b). The metamorphism of these two types of rocks may be closely related to the hydrothermal process caused by the activity of the F_{17} fault.

In this paper, the sampling and subsequent research are aimed at the ultramafic rock, red-coloured altered rock and schist.

The ultramafic rock was sampled at the II and IV mineral areas, which are two sides of the F_{17} fault (Figure 2). The sample is dark green in colour, with a fine-grained texture and massive structure, in which serpentinization can be observed. Observed under the microscope, the lithology is lherzolite, and the mineral associations are mainly olivine (80%), orthopyroxene (10%), and clinopyroxene (10%; Figure 3c) with different degrees of serpentinization.

The red-coloured altered rock was sampled at the F_{17} fault zone (Figure 2), and gradually transitions into the ultramafic rock without a clear boundary. The sample is red to brown in colour, with a cryptocrystalline texture, massive structure, and serpentinization can be observed. Observed under the microscope, veinlets and stockwork crosscut into the red brown matrix, and the matrix minerals are not visible (Figure 3d). The X-ray powder diffraction test revealed that the sample is composed of iron oxide, such as haematite, specularite, magnetite, and some calcite and serpentine (Figure 3e). We suggest that the red-coloured altered rock was metamorphosed from the ultramafic rock.

The schist was sampled from the F_{17} fault fracture zone (Figure 2). Underlying the Baijiazui Formation, migmatite is a faulted contact with the NW side of the red-coloured altered rock on the northwest side and a diorite intrusion on the southeast side (Figure 3b,c). The sample is grey to green in colour with a schistose structure. Observed under the microscope, the lithology is chlorite tremolite schist, and the

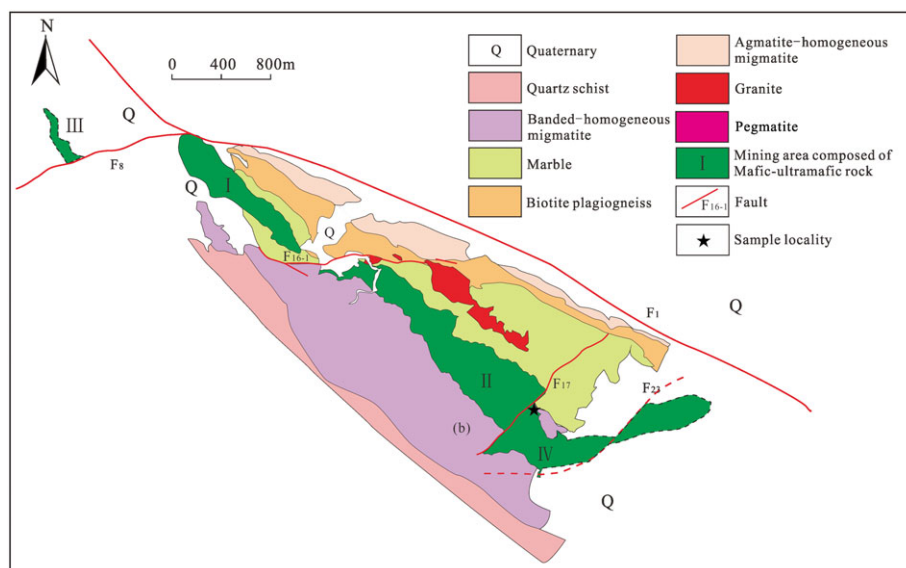


FIGURE 2 Geological map of Jinchuan mineral area (after Zeng, 2016) [Colour figure can be viewed at wileyonlinelibrary.com]

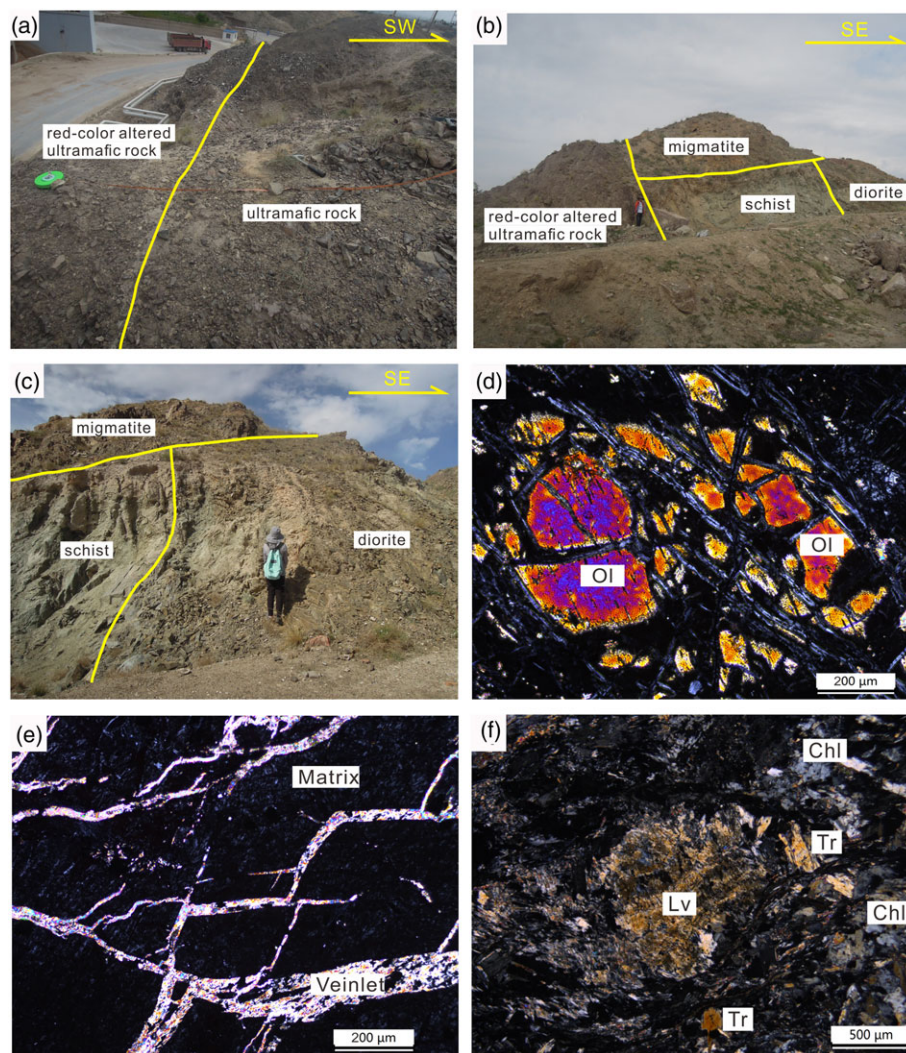


FIGURE 3 Field outcrop and photomicrographs (in cross-polarized light) of the samples in Jinchuan mineral area: (a) contact zone of ultramafic rock and red-colour altered rock; (b) contact zone of red-colour altered rock, migmatite, and schist; (c) contact zone of migmatite, schist, and diorite; (d) photomicrograph of ultramafic rock; (e) photomicrograph of red-colour altered rock; (f) photomicrograph of schist. Ol: olivine; Chl: chlorite; Tr: tremolite; Lv: lithic fragment [Colour figure can be viewed at wileyonlinelibrary.com]

mineral association is mainly chlorite (30%) and tremolite (70%; Figure 3f). We suggest that the chlorite tremolite schist is a metamorphosed fault clastic.

4 | ANALYTICAL METHODS

4.1 | X-ray powder diffraction

A small amount powder of red-coloured altered rock was selected for the X-ray powder diffraction. The X-ray powder diffraction was analysed at the Laboratory of X-ray Diffraction, Central South University, using a Japanese-made D/max-2500 high-powered X-ray powder diffractometer. The test conditions are as follows: The X-ray uses a copper target with 40 KV voltage and 200 mA of electricity. The wave filtering uses a graphite monochromator with $SS = DS = 1^\circ$ and $RS = 0.05$ mm slit. The detector uses a scintillation counter with continuous scanning, a 0.02° increment, $10^\circ/\text{min}$ scanning speed, a $5\text{--}56^\circ$ scanned area, and 0.000° θ deviation. The results are shown in Figure 4.

4.2 | Zircon U-Pb dating and trace elements

More than 50 kg of red-coloured altered rock sample and one schist sample were selected for the zircon U-Pb dating and trace elements analysis. Zircon separation was performed at Langfang Integrity Geological Services Co. Ltd using the single mineral conventional separation method. Zircon target and cathodoluminescence (CL) images were obtained from Chongqing Yujing Science and Technology Services Co. Ltd using gum to attach the zircons to the target, and a scanning electron microscope for the CL photographs. Zircon LA-ICP-MS U-Pb dating and trace elements were performed at the Nanjing FocuMS Technology Co. Ltd using an inductively coupled plasma mass spectrometer (LA-ICP-MS, Agilent 7700 x) equipped with a Microlas system (Analyte Excite). The beam spot is $35\text{ }\mu\text{m}$, energy density is $6.0\text{ J}/\text{cm}^2$, it has a 4 Hz laser pulse repetition rate, and denudation time is 40 s. The zircon 91500 and NIST SRM 610 were used as the external standards for the U-Pb isotope ratios and element contents, respectively (Liu, Hu, Zong, Gao, & Gao, 2010). For detailed instrument parameters and the analysis process, refer to Sun et al. (2013).

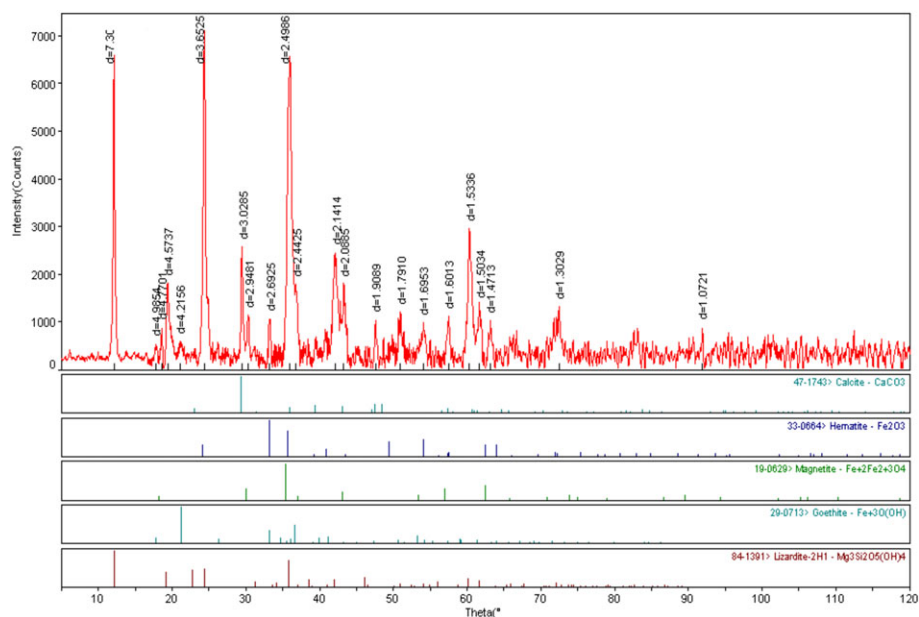


FIGURE 4 X-ray powder diffraction result of red-colour altered rock [Colour figure can be viewed at wileyonlinelibrary.com]

and Gu et al. (2013). Processing software for the U/Pb is LaDating (version 1.5), quantitative calibration for Pb isotope dating was performed by ComPbcorr#3_18 (Andersen, 2002), and concordia diagrams and weighted mean calculations were performed using ISOPLOT 4.15 (Ludwig, 2003).

4.3 | Zircon Lu-Hf isotope

Zircons with good concordance from the red-coloured altered rock sample and schist sample were selected for the zircon Lu-Hf isotope analysis. Zircon LA-MC-ICP-MS Lu-Hf analyses were measured at the Nanjing FocuMS Technology Co. Ltd, using a GeoLas 193-nm ArF laser and Thermo Scientific Neptune multicollector ICP-MS, and the method is outlined by Kemp et al. (2009). Spot sizes were 60 μm with a 4 Hz laser pulse repetition rate. Ablation was conducted in the He gas flow. The Mud Tank reference zircon and Geostandard FC1 zircon were used as the external standards, and the $^{176}\text{Hf}/^{177}\text{Hf}$ of these two external standards are 0.282184 and 0.282507, respectively (Woodhead & Hergt, 2005). In the experiment, the standard zircon Mud Tank reference zircon and FC1 were analysed, and the $^{176}\text{Hf}/^{177}\text{Hf}$ ratios were 0.282148–0.282175 and 0.282488–0.282502, respectively, which is in accordance with the recommended value.

5 | ANALYTICAL RESULTS

5.1 | Zircon U-Pb ages

A total of thirty zircons of red-coloured altered rock sample JH-16 and over 1,000 zircons from the schist sample JH-38 were separated. We chose thirty zircons from sample JH-16 and 200 zircons from sample JH-38 for U-Pb dating and trace elements analysis. The LA-ICP-MS zircon U-Pb isotopic dating results are listed in Supplementary Table 1.

The thirty effective spots of the 30 zircons were obtained from sample JH-16, and the representative CL image is shown in Figure 5. In the concordia diagrams (Figure 6a), 27 spots are concordant (concordant ratio > 90%). In the zircon relative probability diagram

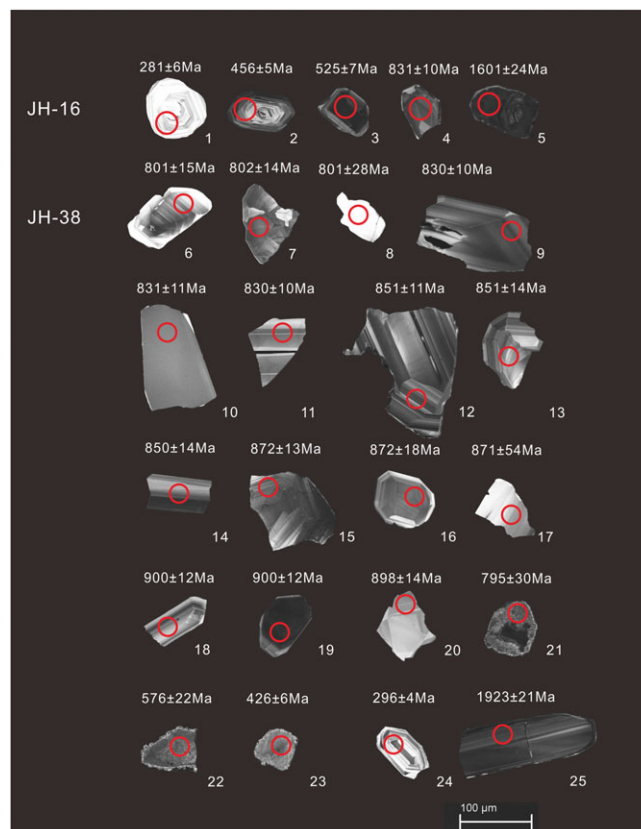


FIGURE 5 Representative cathodoluminescence image of zircons from sample JH-16 and JH-38 [Colour figure can be viewed at wileyonlinelibrary.com]

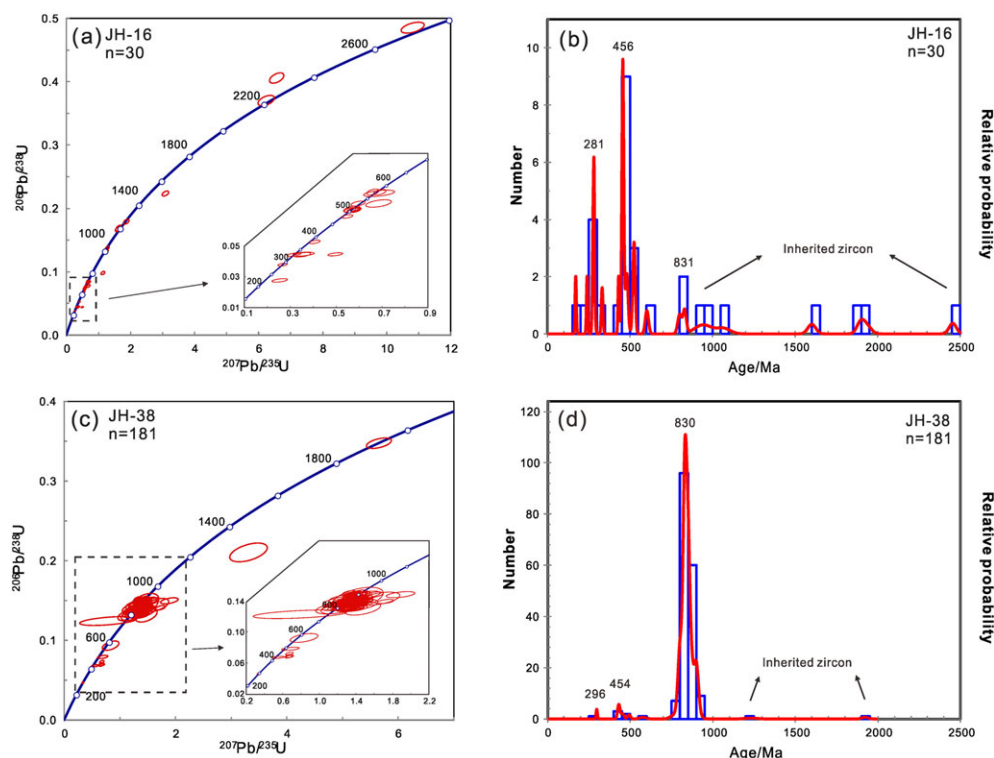


FIGURE 6 Zircon U-Pb concordia diagrams and relative probability diagrams of zircons from sample JH-16 and JH-38 [Colour figure can be viewed at wileyonlinelibrary.com]

(Figure 6b), seven spots yield $^{206}\text{Pb}/^{238}\text{U}$ ages older than 1,000 Ma with $^{207}\text{Pb}/^{206}\text{Pb}$ ages of 934; 951; 1,065; 1,601; 1,896; 1,932; and 2,455 Ma, which are notably older than the ~830 Ma ore-bearing ultramafic rock and represent an inherited zircon. There were 20 spots that yielded $^{206}\text{Pb}/^{238}\text{U}$ ages younger than 1,000 Ma, which can be divided into four age groups: 831–830 Ma in two spots, 526–521 in three spots, 481–431 in 10 spots, and 331–240 in five spots. Those that are 831–801 Ma are adjacent to the formation of the Jinchuan ore-bearing ultramafic rock, and the 526–521, 481–431, and 331–240 Ma ages may represent hydrothermal activities that occurred during a later period.

There were 181 effective spots in the 181 zircons obtained from sample JH-38, and the representative CL image is shown in Figure 5. In the concordia diagrams (Figure 6c), 167 spots are concordant (concordant ratio > 90%). In the zircon relative probability diagram (Figure 6d), one spot yields a $^{206}\text{Pb}/^{238}\text{U}$ age older than 1,000 Ma with a $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1,906 Ma, which is notably older than other zircons and represents an inherited zircon. A total of 167 spots yield $^{206}\text{Pb}/^{238}\text{U}$ ages younger than 1,000 Ma, which can be divided into five age groups: 910–897 Ma in 17 spots, with a weighted mean of 900.4 ± 6.2 Ma (Figure 7a, 95% confidence, MSWD = 0.065); 872–869 Ma in 15 spots, with a weighted mean of 870.6 ± 7.1 Ma (Figure 7b, 95% confidence, MSWD = 0.0058); 852–849 Ma in 40 spots, with a weighted mean of 850.7 ± 3.6 Ma (Figure 7c, 95% confidence, MSWD = 0.0055); 832–827 Ma in 73 spots, with a weighted mean of 830.8 ± 2.7 Ma (Figure 7d, 95% confidence, MSWD = 0.0077); and 802–797 Ma in 20 spots, with a weighted mean of 800.5 ± 5.5 Ma (Figure 7e, 95% confidence, MSWD = 0.019). Except for the five

groups, two single spots can be distinguished with $^{206}\text{Pb}/^{238}\text{U}$ ages of 576 and 296 Ma, respectively. There are eight spongy structure zircons in sample JH-38, seven of which are discordant, with a concordant ratio of approximately 82%. One concordant spot yielded a $^{206}\text{Pb}/^{238}\text{U}$ age of 576 Ma, and another seven discordant spots yielded $^{206}\text{Pb}/^{238}\text{U}$ ages of 795 and 492–426 Ma.

5.2 | Zircon trace elements

The trace elements analysis was performed simultaneously with the U-Pb analysis, and the obtained elements were P, Ca, Ti, Rb, Sr, Nb, Ta, Hf, Y, Pb, Th, U, and REE; the SiO_2 content is used for correction. The results are listed in Supplementary Table 2.

A total of 19 zircons from sample JH-16 with $^{206}\text{Pb}/^{238}\text{U}$ ages younger than 1,000 Ma are selected for the chondrite-normalized REE diagram and discrimination. The chondrite-normalized REE patterns are shown in Figure 8a. The REE patterns can be divided into two types (expressed as Types 1 and 2 in the following): Type 1 has an REE content of between 705.70 and 1,655.18 $\mu\text{g/g}$, with an intensely positive Ce anomaly and negative Eu anomaly; Type 2 has an REE content of between 840.10 and 2,932.98 $\mu\text{g/g}$, which is an obviously high and invariable LREE content, with a slightly positive Ce anomaly and negative Eu anomaly.

There are two zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages of ~900, ~870, ~850, ~830, ~800, and seven zircons with spongy structures from sample JH-38 are selected for the chondrite-normalized REE diagram and discrimination. The chondrite-normalized REE patterns are shown in Figure 8b, and the REE patterns can be divided into two types

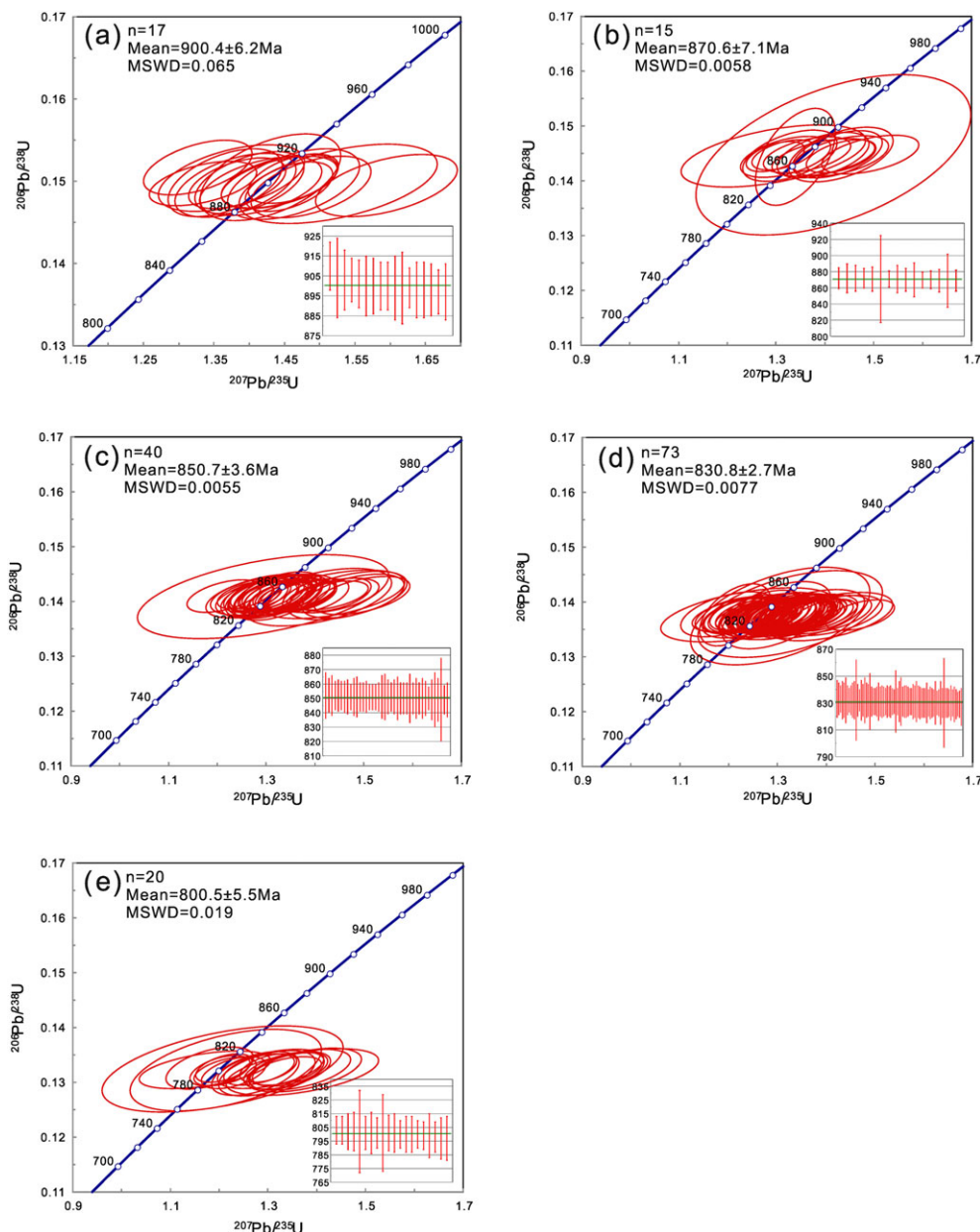


FIGURE 7 Concordia diagrams and weighted mean age diagrams of zircons from sample JH-38. (a) zircons with ages about ~900 Ma, (b) zircons with ages about ~870 Ma, (c) zircons with ages about ~850 Ma, (d) zircons with ages about ~830 Ma, and (e) zircons with ages about ~800 Ma [Colour figure can be viewed at wileyonlinelibrary.com]

(expressed as Types 3 and 4 in the following): Type 3 has an REE content between 952.44 and 2,844.63 $\mu\text{g/g}$, with an intensely positive Ce anomaly and negative Eu anomaly; Type 4 has an REE content of between 1,398.82 and 3,300.58 $\mu\text{g/g}$, with flat REE patterns and almost no Ce or Eu anomalies.

5.3 | Zircon Lu-Hf isotope

Zircons of different age groups with high concordant ratios were selected for the Hf isotope analysis. The results are listed in Supplementary Table 3.

Ten zircons from sample JH-16 were selected for Hf isotope analysis, which include nine Type 1 zircons and one Type 2 zircon grain. The $^{176}\text{Hf}/^{177}\text{Hf}$ values are between 0.282149 and 0.282416,

$^{206}\text{Pb}/^{238}\text{U}$ ages are between 831 and 456 Ma, and the $\epsilon\text{Hf}(t)$ are between -7.01 and -2.80 . The one stage model age (T_{DM}) is between 1,468 and 1,178 Ma, and the two-stage model age (T_{DM2}) is between 1,870 and 1,594 Ma. In the diagram of zircon $\epsilon\text{Hf}(t)$ values vs. ages (Figure 9), the zircons of sample JH-16 are all below the CHUR evolution line.

A total of 66 zircons from sample JH-38 were selected for Hf isotope analysis, which include fifty-nine Type 3 zircons and seven Type 4 zircons. The $^{176}\text{Hf}/^{177}\text{Hf}$ values are between 0.282079 and 0.282219, $^{206}\text{Pb}/^{238}\text{U}$ ages are between 904 and 426 Ma, and the $\epsilon\text{Hf}(t)$ are between -14.82 and -2.47 . The one-stage model age (T_{DM}) is between 1,651 and 1,341 Ma, and the two-stage model age (T_{DM2}) is between 2,046 and 1,863 Ma. In the diagram of zircon $\epsilon\text{Hf}(t)$ values vs. ages (Figure 9), the zircons of sample JH-38 are all below the

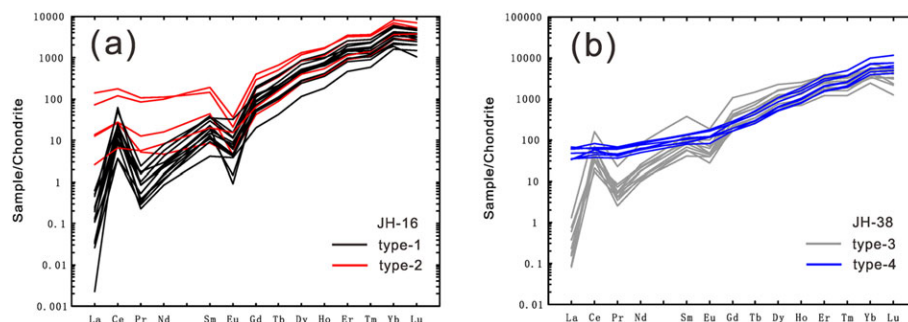


FIGURE 8 Chondrite-normalized REE diagrams of representative zircons from sample JH-16 and JH-38 (normalized after Sun & McDonough, 1989) [Colour figure can be viewed at wileyonlinelibrary.com]

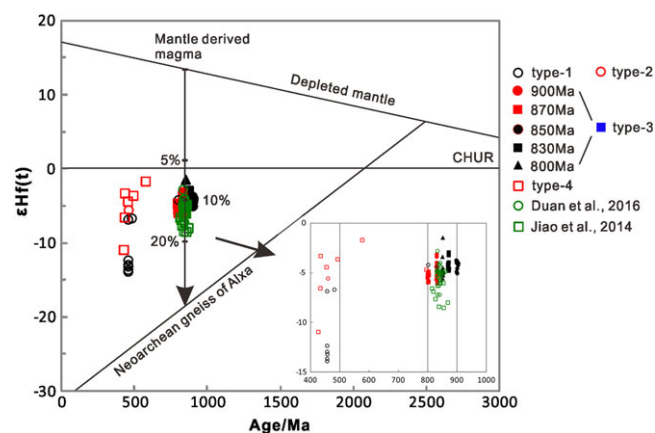


FIGURE 9 Zircon $\epsilon_{\text{Hf}}(t)$ values vs. ages diagram. Neoproterozoic gneiss data are from Gong, Zhang, Yu, Li, and Hou (2012); mixing calculating parameters are from Rudnick and Gao (2003) and Sun and McDonough (1989): upper crust Hf average content 7.8 ppm, $\epsilon_{\text{Hf}}(t) = -19$; mantle Hf average content 0.8 ppm, $\epsilon_{\text{Hf}}(t) = 14$ [Colour figure can be viewed at wileyonlinelibrary.com]

CHUR evolution line. Among the Type 3 zircons, the 850–830 Ma zircons showed U–Pb ages and $\epsilon_{\text{Hf}}(t)$ values very similar to the zircons of the Jinchuan ore-bearing ultramafic rock reported by Duan et al., (2016) and Jiao et al. (2014).

6 | DISCUSSION

6.1 | Genesis of the zircons

6.1.1 | Discriminating from morphology

Zircons with different genesis and host rocks usually have different internal structures. The CL images of igneous zircons usually have oscillatory zoning, and the width of oscillatory zoning may be related to the crystallization temperature of the magma. Wide oscillatory zoning typically formed at a high temperature condition because of fast element diffusion. Conversely, narrow oscillatory zoning usually formed at a low temperature condition because of slow element diffusion. Many zircons from mantle rocks show no oscillatory zoning compared to crustal rocks (Wu & Zheng, 2004).

Zircons related to hydrothermal activities usually have special and various internal structures. Zircons crystallized from hydrothermal

fluid usually have a regular shape, with obvious banded zoning or oscillatory zoning. Zircons recrystallized in hydrothermal fluid usually have a porous or spongy structure, which is rich in fluid and mineral inclusions. The completely recrystallized zircons will show a skeletal structure due to the dissolution of hydrothermal fluid (Wu & Zheng, 2004).

Zircons from sample JH-16 are light yellow or transparent, with a stumpy to elongated columnar shape, sizes between 80 and 150 μm , subhedral to euhedral in shape, and they have different characteristics in different age groups, which is described as follows:

Zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages older than 1,000 Ma are probably inherited; they are dark in the CL image without oscillatory zoning, and some of them have a bright core. These zircons may represent the metamorphic zircons captured in the old strata; some may have undergone later metamorphism, as they exhibit an obvious core–mantle structure (Figure 5, No. 5).

The zircon grain with a $^{206}\text{Pb}/^{238}\text{U}$ age of 831 Ma has obviously oscillatory zoning and an inherited core. The zircon grain with a $^{206}\text{Pb}/^{238}\text{U}$ age of 801 Ma does not have oscillatory zoning, but it does have banded zoning (Figure 5, No. 4). These two grains are most likely morphologically metamorphic zircons instead of original magmatic zircons.

Zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages younger than 800 Ma are euhedral in shape, with stumpy to elongated columnar shapes, and they have obvious oscillatory zoning (Figure 5, Nos. 1, 2, and 3), which may represent zircons that crystallized from later hydrothermal fluid.

Zircons from sample JH-38 are light yellow or transparent. The grains are varied in size and morphology. The sizes are between 50 and 200 μm with anhedral or cracked shapes. They also have different characteristics in different age groups, which are described as follows:

The zircon grains with $^{206}\text{Pb}/^{207}\text{Pb}$ age of 1,906 Ma are elongated, columnar in shape with banded zoning, which is similar to the zircons in the metamorphic vein (Figure 5, No. 25). The zircon grain with a $^{206}\text{Pb}/^{238}\text{U}$ age of 296 Ma is stumpy, columnar in shape with bright oscillatory zoning, which is similar to the zircons that crystallized from the hydrothermal fluid (Figure 5, No. 24).

The zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages of 900–800 Ma exhibit various and complicated morphologies. These zircons are varied in size with different structures, such as oscillatory zoning, banded zoning, sector zoning, or no zoning (Figure 5, No. 6–20). However, the grains are mostly very cracked with a 100 Ma age gap, which represents multiple origins of the fault fracture zone.

In addition, some spongy structured or skeletal structured zircons exist in sample JH-38 (Figure 5, Nos. 21, 22, and 23). These zircons are all discordant and varied in ages, which represents a transform or recrystallization by later hydrothermal fluid.

6.1.2 | Discriminating from trace elements

Trace elements can be used to effectively discriminate zircon genesis, as well as the host rocks. The petrography we mentioned above indicates that the sample JH-16, which we referred to as a red altered rock, is a product of oxidated ultramafic rock. Sample JH-38, which we referred to as chlorite tremolite schist, is exposed in a fault fracture zone and may be a product of metamorphic fault clastics. The ultramafic rocks are not saturated in Zr; thus, it is difficult for ultramafic rock to crystallize out the zircons. The zircons from sample JH-16 may not represent an original magmatic zircon but a captured zircon or hydrothermal zircon. Zircons from sample JH-38 with ages of 900–800 Ma may represent different phases of magmatic events, and the zircons with spongy structures indicate several later-occurring hydrothermal fluid events. Therefore, the sample JH-16 may be a metamorphic product of ore-bearing ultramafic rock, but the sample JH-38 may not, which is different from the opinion that the red altered rock and chlorite tremolite schist are products from the metamorphic ore-bearing ultramafic rock (Zeng, Lai, Mao, Ai, & Yue, 2015). We do not discuss the zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages older than 1,000 Ma, as these grains may represent the captured zircons from the old strata and are unrelated to the magmatic events that occurred during the Neoproterozoic.

As mentioned above, the zircons from sample JH-16 can be divided into two types, according to the REE pattern, and expressed as Types 1 and 2. The zircons from sample JH-38 can be divided into two types, according to the REE pattern, which are expressed as Types 3 and 4. We use trace elements to discriminate the four types as follows.

Hoskin (2005) considered that the hydrothermal zircon contains much more REE content and exhibits a notably different REE pattern compared with the magmatic zircons. Hydrothermal zircons usually have more negative Ce anomalies and a high LREE content, due to containing LREE abundant mineral inclusions and fluid inclusions (Fu, Mernagh, Kita, Kemp, & Valley, 2009; Li, Watanabe, & Yonezu, 2014; Pettke, Aude'ta, Schaltegger, & Heinrich, 2005; Schaltegger, Pettke, Aude'ta, Reusser, & Heinrich, 2005). Types 2 and 4 zircons have more negative Ce anomalies and high LREE contents compared with the Types 1 and 3 zircons (Figures 8 and 10a,b), which indicates their hydrothermal origin. Because the hydrothermal zircons have a high LREE content, their $(\text{Sm}/\text{La})_{\text{N}}$ content will be lower. Hoskin (2005) proposed the La vs. $(\text{Sm}/\text{La})_{\text{N}}$ and $(\text{Sm}/\text{La})_{\text{N}}$ vs. Ce/Ce^* diagrams to discriminate the hydrothermal zircon and magmatic zircon. In these two diagrams, Types 2 and 4 zircons are all plotted in a hydrothermal zircon area (Figure 10c,d).

The Th and U contents and Th/U value are effective geochemistry indicators for zircon discrimination, because magmatic zircon usually have high Th and U contents and a Th/U value greater than 0.5; metamorphic zircon have low Th and U contents and a Th/U value lower than 0.01 (Guo, Zhang, Harris, Xu, & Pan, 2016, 2017; Hoskin &

Schaltegger, 2003). However, the Th and U contents and Th/U value of hydrothermal zircon are varied. In the diagrams of U vs. Th and Th/U vs. Hf, Type 3 zircons have a high Th and U content, with a $\text{Th}/\text{U} > 0.5$. Types 1 and 4 zircons have relatively high Th content but low U content, with a $\text{Th}/\text{U} < 0.5$. Type 2 zircons have various Th and U content, with a $\text{Th}/\text{U} < 0.5$ (Figure 10e,f).

In conclusion, we consider that the Types 2 and 4 zircons have characteristics of hydrothermal zircons, whereas the Type 3 zircons have characteristics of magmatic zircons. Type 1 zircons have REE patterns similar to the magmatic zircon, but the Th and U contents and Th/U value are much lower than the magmatic zircon, with ages younger than the formation age of the Jinchuan ore-bearing ultramafic rock. Thus, we consider them to be metamorphic zircons, and the ages reflect metamorphism ages from after magmatic crystallization. Interestingly, these metamorphism ages correspond with the hydrothermal zircon ages of Type 4. Thus, we believe that the F_{17} fault zone may have experienced several metamorphic hydrothermal events, with the greatest amount of zircon recorded at 450–430 Ma, which indicates the largest scale hydrothermal event.

6.1.3 | Zircon host rock discrimination

As mentioned above, Type 3 zircons have characteristics of magmatic zircon ($\text{Th}/\text{U} > 0.5$, depletion in LREE content, and intensely negative Ce anomaly). Despite the chlorite tremolite schist lithology of sample JH-38, the zircons have different ages between 900 and 800 Ma, which indicates multiple origins. Thus, the chlorite tremolite schist is exposed in a fault fracture zone and combined with the cracked structure of the zircons; we suggest that the chlorite tremolite schist may be derived from the metamorphism of fault clastics with multiple origins.

Type 3 zircons have $^{206}\text{Pb}/^{238}\text{U}$ ages between 900 and 800 Ma and can be divided into five age groups, which indicates multiple origins. This makes it necessary to discriminate the host rocks. Because trace elements of few zircons are under the detection limit, we only choose one 163 Type 3 zircons for their host rock discriminations as follows.

The statistics from Belousova, Griffin, O'Reilly, and Fisher (2002) show trace elements content in zircons from multiple kinds of rocks, which indicates that zircons from different host rocks have prominently different trace elements contents; this is shown by the rapid REE content decrease from acid rocks to ultramafic rocks. Type 3 zircons have REE contents between 1,000 and 3,000 ppm. Compared with the statistical data, these zircons are much lower than the REE content of zircons from granitoids ($>10,000$ ppm), only slightly higher than the REE contents of zircons from lamproites (600–700 ppm) and close to the REE contents of zircons from basalts and dolerites (1,000–2,000 ppm).

Belousova et al. (2002) considered that incompatible elements tend to be enriched in rocks with a high degree of differential evolution. As representative incompatible elements, Th and U contents have an increasing trend from mafic rocks to acid rocks in the zircon grains. The Th^{4+} and U^{4+} can replace the Zr^{4+} , and thus, the Th and U contents have the same trend in zircons when compared with the whole rock and also increase from mafic rocks to acid rocks.

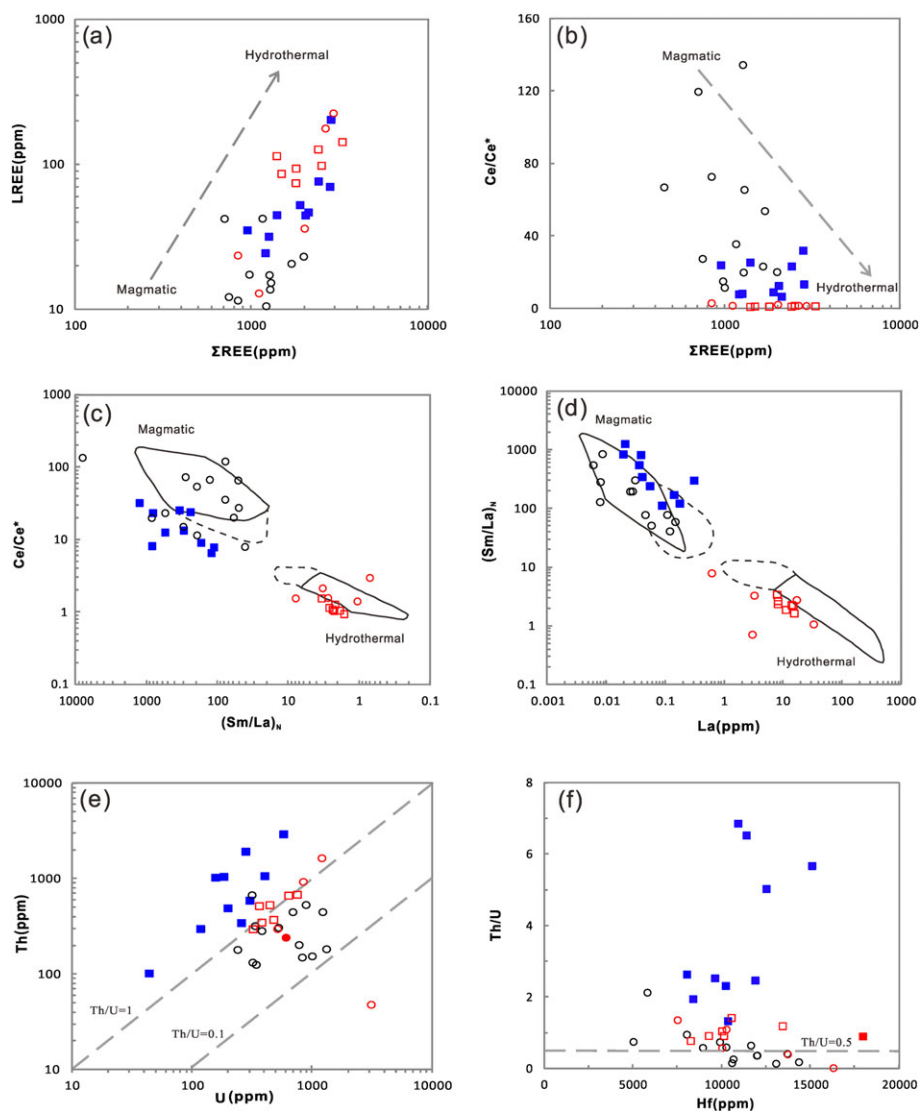


FIGURE 10 Discriminant diagram of (a) ΣREE vs. LREE, (b) ΣREE vs. Ce/Ce^* , (c) $(\text{Sm}/\text{La})_N$ vs. Ce/Ce^* , (d) La vs. $(\text{Sm}/\text{La})_N$, (e) U vs. Th, and (f) Hf vs. Th/U for the zircons from sample JH-16 and JH-38 (Graphic symbols are in Figure 9) [Colour figure can be viewed at wileyonlinelibrary.com]

Similarly, Nb, Ta, REE, and Y have the same geochemical behaviours, but they have different capabilities to replace Zr^{4+} , due to their different ionic radii. HREE^{4+} , U^{4+} , and Ta^{4+} have ionic radii closer to Zr^{4+} than LREE^{4+} , Th^{4+} , and Nb^{4+} , and thus, the former are easier to access in the zircons (Ahrens, Cherry, & Erlank, 1967). However, almost all magmatic originating zircons have a high Th/U value. This is opposite from the fact that U^{4+} is easier to access in the zircons than Th^{4+} . Belousova et al. (2002) suggested it is because the host rock magma is enriched in Th contents and depleted in U contents, as well as enriched in Nb contents and depleted in Ta contents. Thus, most magmatic zircons have the characteristics of high Th and Nb but low U and Ta contents, which is a characteristic of the host rock magma.

Ce usually appears as Ce^{3+} but will be oxidized to Ce^{4+} in an oxidation environment. The $\text{Ce}^{3+}/\text{Ce}^{4+}$ values are typically used to measure the degree of oxidation for the whole rock. In zircons, Ce^{4+} has an ionic radius closer to Zr^{4+} than Ce^{3+} . Thus, it is easier for Ce^{4+} to replace Zr^{4+} and access zircons, and therefore, the zircons from a rock with a higher oxidation degree usually have an intensely positive Ce anomaly. Eu appears as Eu^{2+} and Eu^{3+} , Eu^{2+} is less compatible than

Eu^{3+} , but Eu^{2+} easily replaces Ca^{2+} and accesses feldspar grains; during the fractional crystallization or partial melting process, the separation of feldspar will cause negative Eu anomaly in the magma; this will also reflect in the zircons. Thus, zircons from a rock with a high degree of fractional crystallization or experienced low levels of partial melting will have an intensely negative Eu anomaly, due to the Eu separation with feldspars and Eu absence in the magma.

According to this, Belousova et al. (2002) proposed a series of discriminant diagrams to estimate the host rock type of the zircons as follows (Figure 11a–f). In the diagram of U vs. Y, Eu/Eu^* vs. Ce/Ce^* , $(\text{Yb}/\text{Sm})_N$ vs. Y, Nb/Ta vs. Y, Ce/Ce^* vs. Y, and Ta vs. Nb, the Type 3 zircons have incompatible elemental contents similar to the intermediate-basic rocks, but completely different from the granitoid, kimberlite, and carbonatite, and a high degree of oxidation and Ce/Ce^* value. Although the area of different host rocks has overlaps, most of the Type 3 zircons are plotted in the area of larvikite, which indicates an alkaline-intermediate lithology.

Belousova et al. (2002) also proposed a CART (classification and regression tree) way of using the trace elements of zircons to

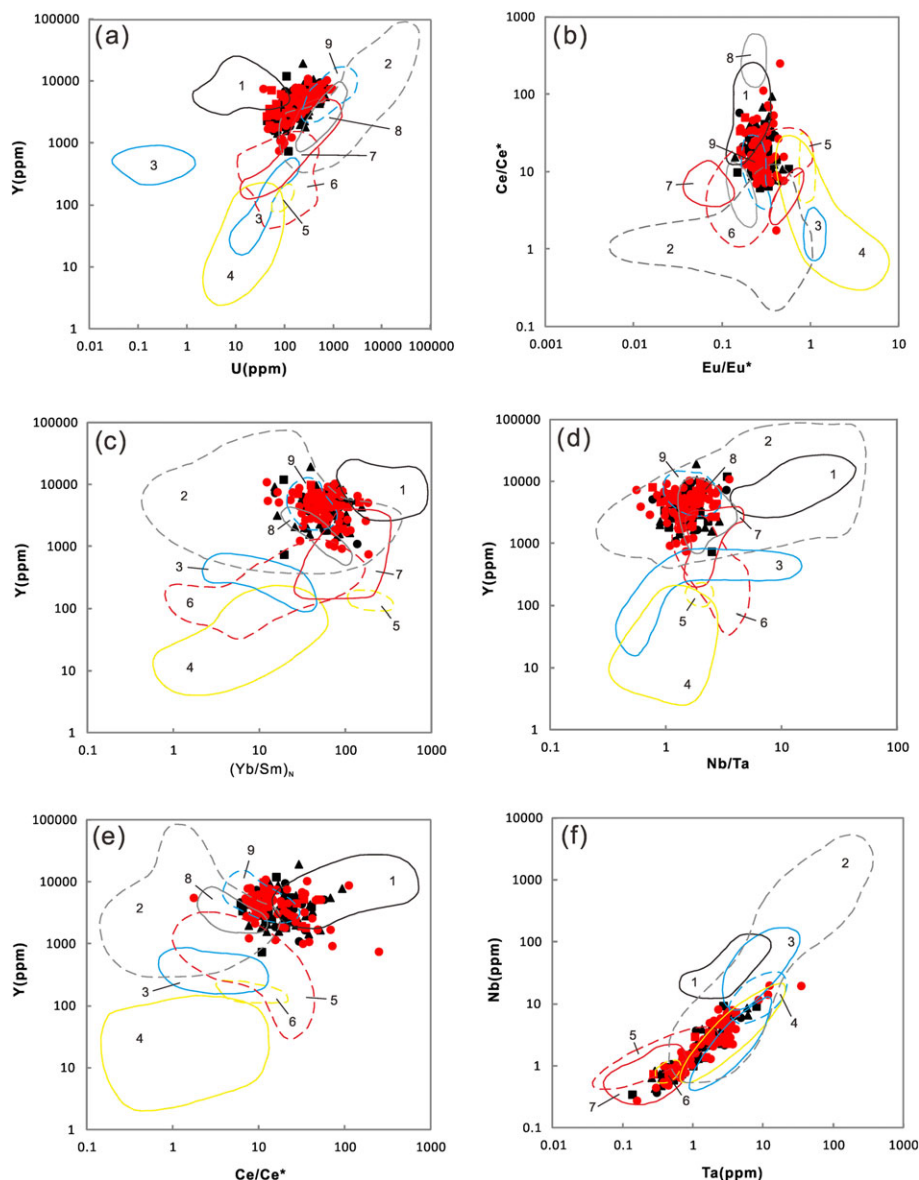


FIGURE 11 Host rock discrimination diagrams of magmatic zircons (after Belousova et al., 2002). The numbers of the fields represent (1) nepheline syenite and syenite pegmatite; (2) granitoid; (3) carbonate; (4) kimberlite; (5) syenite; (6) lamproite; (7) mafic rock; (8) syenite pegmatite; and (9) larvikite. (Graphic symbols are in Figure 9) [Colour figure can be viewed at wileyonlinelibrary.com]

discriminate the host rock type (Figure 12a). This is used to discriminate all Type 3 zircons ($n = 163$) and illuminate multiple host rock types with larvikite ($n = 131$), dolerite ($n = 30$), Ne-Syenite ($n = 1$), and granitoid ($n = 1$; Figure 12b). However, a short CART he proposed was not used here due to the inaccurate comparison with the CART.

Combined with the zircon morphology, the Type 3 zircons are mostly small grains (50–150 μm) with no oscillatory zoning, which is similar to the zircons from mafic rocks. In conclusion, we consider the discrimination above to be effective, and the Type 3 zircons mostly originate from intermediate-basic rocks.

6.2 | Origin of the zircons

6.2.1 | Source of the zircon host rocks

The Type 3 zircons have $^{206}\text{Pb}/^{238}\text{U}$ ages between 900 and 800 Ma and can be divided into five groups, but there are no distinct

differences in the trace elements content and $\epsilon\text{Hf}(t)$ values between groups. The Type 3 zircons have $\epsilon\text{Hf}(t)$ values between -2 and -7 , which are distinctly lower than the depleted mantle value, but all have the intermediate-basic host rock, which indicates a crustal material contamination. However, zircons with $\epsilon\text{Hf}(t)$ values slightly lower than zero may have two factors: The first factor is the magma mixing between depleted mantle and crustal origin; another factor is that the magma have a direct source from the enriched mantle.

However, the Type 3 zircons have $\epsilon\text{Hf}(t)$ values with slight variations and without adjacent those of the depleted mantle or ancient crust. If the zircons have a genesis factor of magma mixing, $\epsilon\text{Hf}(t)$ values with a slight variation must have two approximate end member contents. Thus, we propose a mixing model for the Type 3 zircons: The mantle end member is chosen from the 830 Ma depleted mantle, $\epsilon\text{Hf}(t) = 14$; the crust end member is chosen from the Paleoproterozoic gneiss of the AB, $\epsilon\text{Hf}(t) = -19$. To reach a mixing result of $\epsilon\text{Hf}(t) = -2$ to -7 ,

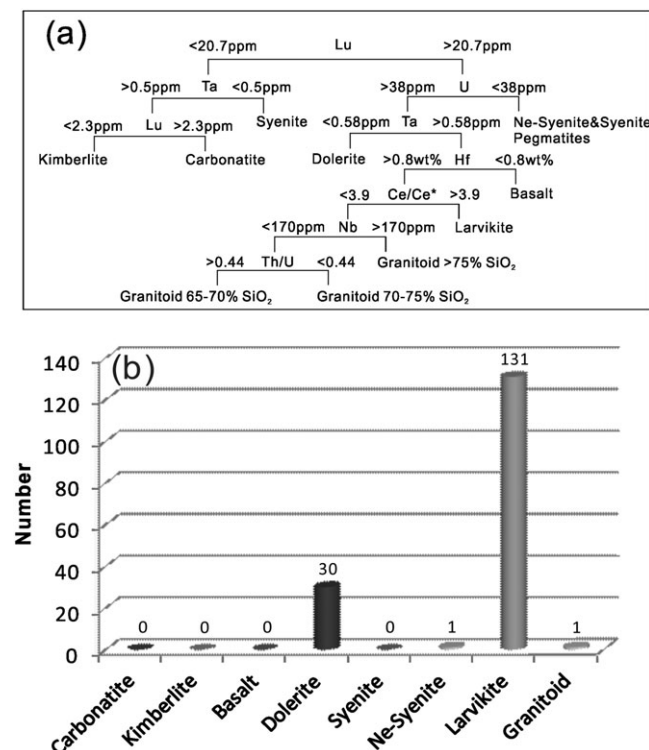


FIGURE 12 CART diagram for (a) zircon host rock discrimination (after Belousova et al., 2002) and (b) host rock statistics

the crust end member has to reach 7–14% (Figure 9), which indicates mantle dominant, and this is inconsistent with the two approximate end member contents. Moreover, the T_{DM} and T_{DM2} of Type 3 zircons have wide gaps, which indicate characteristics of secondary melting. The mantle-dominant magma mixing must result in mostly first melting, with slight gaps between T_{DM} and T_{DM2} .

Previous research also discovered that the zircons from the Jinchuan ore-bearing ultramafic rock have $\epsilon_{Hf}(t)$ values between -4 and -7 (Duan et al., 2015) and -5 and -8 (Jiao et al., 2014). This is similar to our Type 3 zircons. Whole-rock geochemistry and Sr-Nd isotopes indicate that Jinchuan ore-bearing ultramafic rocks have an enriched mantle origin (Jiao et al., 2014; Li et al., 2005). In conclusion, we suggest that the Type 3 zircons have an enriched mantle origin and possibly have the same source as the Jinchuan ore-bearing ultramafic rock. However, their host rocks are mainly alkaline-intermediate rock with a high degree of oxidation, which indicates the host rock magma may have experienced a significant degree of crystallization differentiation after the remelting of the enriched mantle.

6.2.2 | Tectonic setting of the host rocks

Previous researchers have discovered a mass of NW-trending mafic-ultramafic dikes in the Longshoushan area, and the zircon U-Pb dating indicates that there are three main magmatic events that occurred ~ 830 , ~ 420 , and ~ 286 Ma (Duan, 2015). Li et al. (2005) and Zeng (2014) studied the whole-rock Sr-Nd isotopes and high-field-strength element content, suggesting that the AB was in an extension-rift environment at ~ 830 Ma, which may reflect the breakup of the RS. However, except for the ~ 830 Ma mafic-ultramafic rocks in the Longshoushan area, few other acid-intermediate rocks were

discovered during the same period, which makes it difficult to constrain the tectonic setting of the AB during Neoproterozoic.

However, abundant magmatic zircons with $^{206}\text{Pb}/^{238}\text{U}$ ages between 900 and 800 Ma were discovered during this study, and these zircons can be used to constrain the tectonic setting. Schultz, Klemm, and Brätz (2006) proposed a way to discriminate the tectonic setting, according to the different trace elements content between zircons from basaltic rocks of the mid-ocean ridge (N-MORB), volcanic arc (VAB), and within-plate (WPB). Although the host rock of Type 3 zircons is mainly larvikite, there are still a small number of zircons from dolerite. In fact, the zircons from dolerite are nearly the same in terms of trace elements content and U-Pb age as the zircons from larvikite. Thus, we can use these to discriminate the tectonic setting and present all Type 3 zircons.

Schultz et al. (2006) considered that zircons from the WPB have a high REE content, especially LREE, which is higher than zircons from VAB and much higher than zircons from N-MORB. Thus, the zircons from WPB have higher Ce and Yb contents. Moreover, zircons from WPB have more weak HREE fractionation, and thus, they have the lowest $(\text{Yb}/\text{Dy})_N$ value.

The Y has similar geochemical characteristics to the HREE, especially Yb and Lu. The Y content shows a decreasing tendency, which is the same as the Lu content from the WPB zircons to the N-MORB zircons. However, the Hf content remains invariable, and thus, the Lu/Hf value tends to decrease from the WPB zircons to the N-MORB zircons.

The Th^{4+} and U^{4+} have ionic radii similar to Gd^{3+} and Er^{3+} , and thus, they have similar geochemical characteristics. Zircons from WPB have higher REE content, and thus, they have higher MREE and HREE contents than VAB and N-MORB, which resulted in high Gd, Er, Th, and U contents. However, the Th/U content tends to decrease from the WPB zircons to the N-MORB zircons, which may reflect the characteristics of the host rock magma.

According to this, Schultz et al. (2006) proposed a series of discriminant diagrams to estimate the host rock tectonic settings as follows (Figure 13a–f). In the diagram of $(\text{Yb}/\text{Dy})_N$ vs. Y, Ce&Y vs. Y, Lu/Hf vs. Y, Th/Gd, Th/U vs. Y, and U vs. Er, the Type 3 zircons have high Y, Ce, Yb, Th, Gd, and Er contents and mostly plot in the area of the WPB type. However, the Type 3 zircons have low Lu and U contents, in the diagram of the Lu/Hf vs. Y, Th/U vs. Y, and U vs. Er, some plots are out of the area, which may reflect the characteristics of low U and Lu content of the host rock magma (Figure 13c,e,f).

In conclusion, the Type 3 zircons have characteristics similar to WPB zircons, which may reflect that the AB was in an extension-rift environment during 900–800 Ma. This may indicate a new viewpoint that the AB experienced a much longer period of intracontinent extension-rift environment, at least 100 Myr.

6.3 | Tectonic affinity and evolution of the AB

For a long time, most researchers believed that the AB is an inherent part of the NCC, according to the similar Archaean basements. However, in recent years, zircon ages proved that the crystalline basement of the AB was formed during the Paleoproterozoic, but not the Archaean (Dan, Li, Guo, Liu, & Wang, 2012). Despite the new found

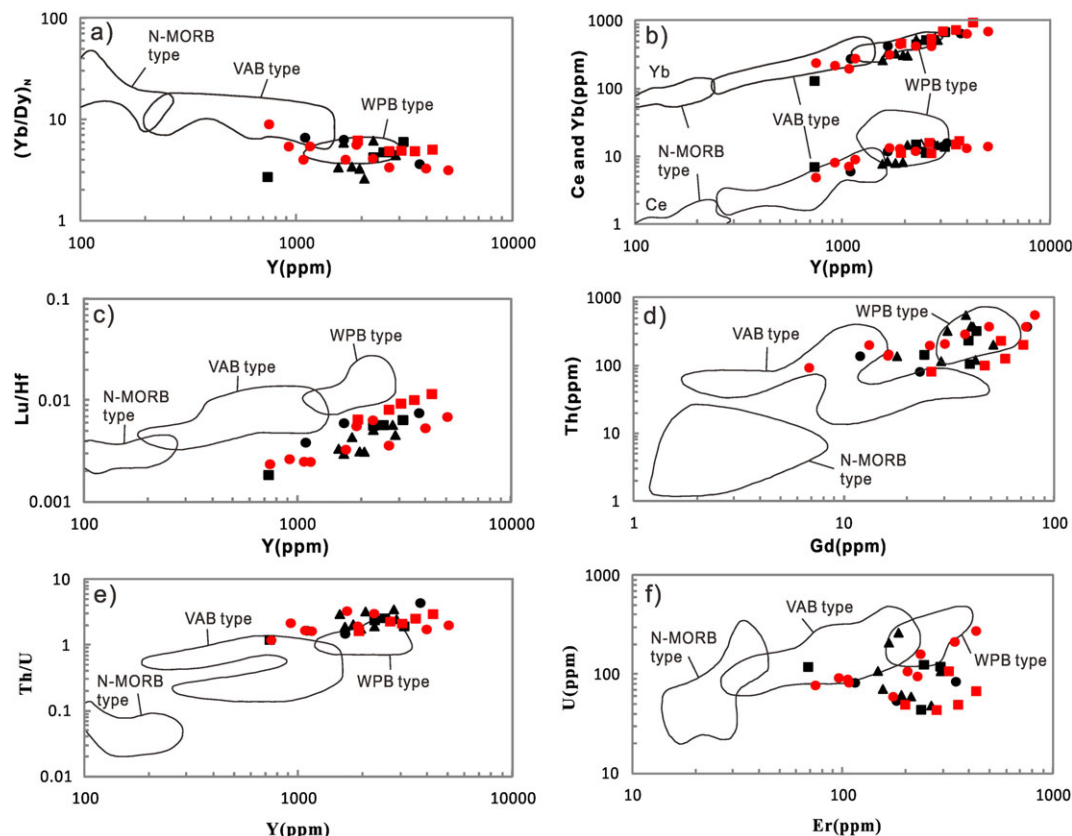


FIGURE 13 Trace element discrimination diagrams for zircons from basic rocks (after Schultz et al., 2006). (Graphic symbols are in Figure 9) [Colour figure can be viewed at wileyonlinelibrary.com]

2.5 Ga granitic gneiss in the western AB, 1.85 Ga Beidashan and Longshoushan migmatization are nearly the same as those of NCC, but this can only prove that the AB have an affinity to the NCC before the Paleoproterozoic and the AB may have participated in the amalgamation of the Columbia supercontinent (Zeng, 2016).

However, an increasing number of Neoproterozoic magmatic events have been discovered in the AB, which are significantly different from the NCC. The detrital zircon data indicate that the age peak in the AB at 2.6–2.4 and 2.1–1.8 Ga are similar to the NCC, but ages of 1.0–0.9 Ga are not found in the NCC. In addition, many ~830 Ma mafic-ultramafic magmatic events were reported in the south-western AB, a 930–910 Ma S-type granitic gneiss was also discovered in the eastern AB (Dan, Li, Wang, Wang, & Liu, 2014). Combined with the 900–800 Ma intermediate-basic magmatic zircons found in this study, we can ensure that the AB underwent abundant magmatic events during the Neoproterozoic, which is rarely reported in the NCC. However, magmatic events during the Neoproterozoic have been widely discovered in the YB, which may be related to the breakup of Rodinia, as some researchers believe that the AB may have been related to the YB during the Neoproterozoic. Accordingly, we use the zircon ages and Hf isotopes, compared with detrital zircons in strata from the AB, western Ordos Block, northern YB and North Qilian Block, to constrain the affinity of the AB.

Zhang, Zhang, Zhang, et al. (2016) and Zhang, Zhang, and Zhao (2016) reported the detrital zircon ages from the Dahuangshan Formation sandstone of the middle Cambrian in the south-western AB and constructed the relative probability diagram shown in Figure 14a,

where there is one major age peak at ~976 Ma and one moderate age peak at ~2,455 Ma. Yuan and Yang (2015) reported the detrital zircon ages from the Shixiagou Formation sandstone of the middle Devonian and the Zhongning Formation sandstone of the upper Devonian in the eastern AB and constructed the relative probability diagram shown in Figure 14b, where there are two major age peaks at ~474 and ~999 Ma, and one moderate age peak at ~2,491 Ma. Zhang et al. (2015) reported the detrital zircon ages from the Xiangshan group sandstone of the middle-upper Cambrian in the south-eastern AB and constructed the relative probability diagram shown in Figure 14c, where there is one major age peak at ~964 Ma, and two moderate age peaks at ~2,517 and ~551 Ma. Darby and Gehrels (2006) reported the detrital zircon ages from the sandstone of strata from the Proterozoic to Ordovician in the western Ordos Block and constructed the relative probability diagram shown in Figure 14d, where there are two major age peaks at ~1,999 and ~2,665 Ma. Wang et al. (2013) reported the detrital zircon ages from the sandstone and schist of strata from the Proterozoic to Cambrian in the northern YB and constructed the relative probability diagram shown in Figure 14e, where there is one major age peak at ~790 Ma; one moderate age peak at ~2,050 Ma; and one minor age peak at ~2,500 Ma. Xu et al. (2010) reported the detrital zircon ages from the Tianzhu Formation and Sijigou Formation sandstone and schist of the upper Ordovician in the North Qilian Orogen and constructed the relative probability diagram shown in Figure 14f, where there is one major age peak at ~480 Ma; one moderate age peak at ~990 Ma; and one minor age peak at ~2,500 Ma.

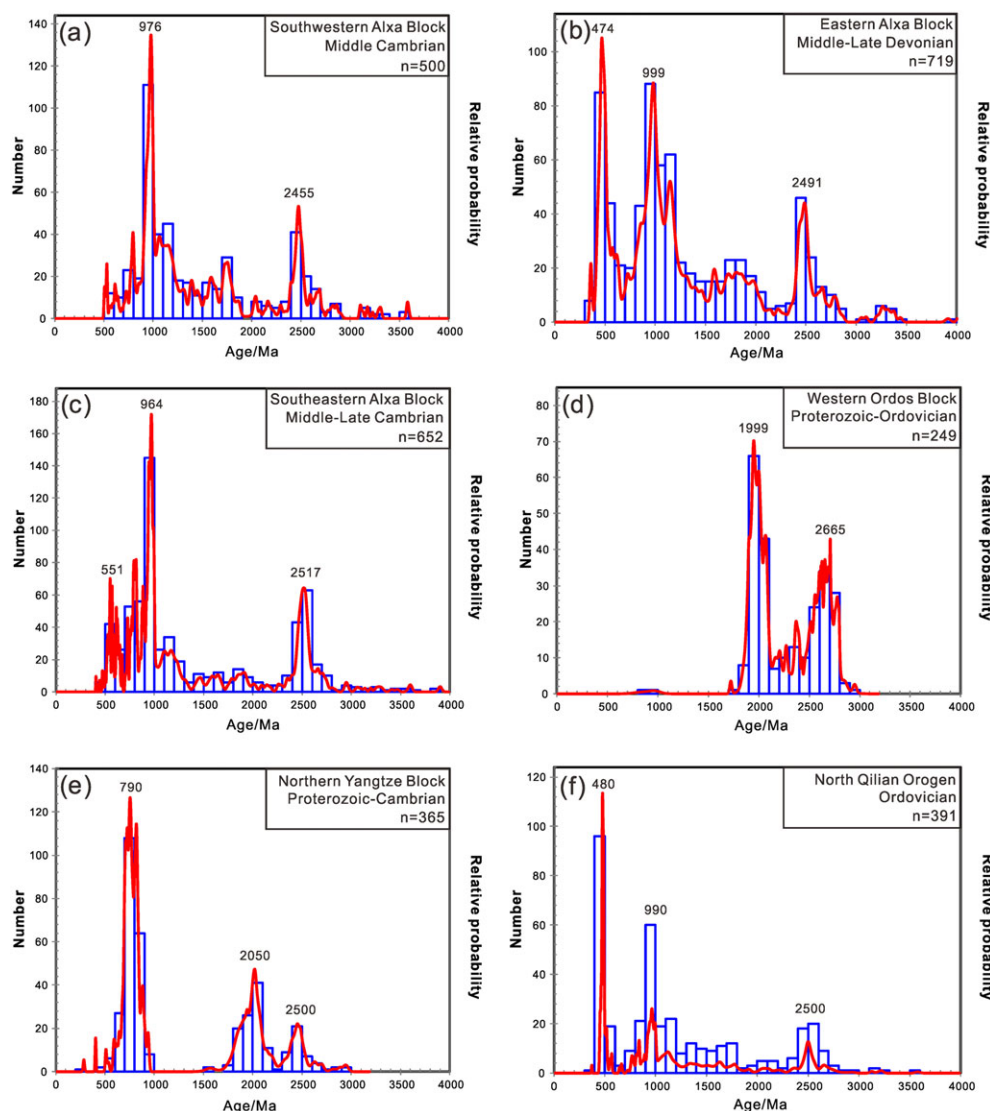


FIGURE 14 Detrital zircon relative probability diagram of (a) the south-western Alxa Block (Zhang, Zhang, Zhang, et al., 2016; Zhang, Zhang, & Zhao, 2016); (b) the eastern Alxa Block (Yuan & Yang, 2015); (c) the south-eastern Alxa Block (Zhang, Zhang, Xiao, Wang, & Zhang, 2015); (d) the western Ordos Block (Darby & Gehrels, 2006); (e) the northern Yangtze Block (Wang, Griffin, Yu, & O'Reilly, 2013); and (f) the North Qilian Orogen (Xu et al., 2010) [Colour figure can be viewed at wileyonlinelibrary.com]

The Ordos Block is in the western part of the NCC, and the detrital zircon data show two major age peaks at ~ 2.0 and ~ 2.6 Ga, which only matches the ~ 2.5 Ga age peak of the AB. Although the detrital zircons from early Paleozoic strata in the AB do not record a ~ 2.0 Ga peak, the detrital zircons from the Precambrian strata recorded the ~ 2.0 and ~ 2.5 Ga age peaks, which are similar to the NCC (Dong et al., 2007). Several ~ 2.0 Ga magmatic events were also discovered in the AB. Dan et al. (2012) discovered abundant ~ 2.0 Ga volcanic rocks in the Bayanwula area, which proved a massive ~ 2.0 Ga magmatic event in the AB. Zeng (2016) and references therein reported an ~ 1.9 Ga meta-mafic rock and 1.9–1.8 Ga migmatization in the Jinchuan mineral area and considered a subduction-collision environment. The ~ 2.0 Ga magmatic event is considered to be related to the amalgamation of the Columbia supercontinent, and the AB and NCC were both took part in this event. Because of the similar ~ 2.0 and ~ 2.5 Ga magmatic events of the AB and NCC, most researchers believe that the AB is an inherent part of the NCC

(Duan, 2015; Gong, 2013); however, an increasing number of discoveries in recent years have indicated that the Neoproterozoic magmatic events are widely developed in the AB, such as the Jinchuan ore-bearing ultramafic rock (Li et al., 2005); the Longshoushan mafic-ultramafic rock (Jiao et al., 2012); S-type granitic gneiss in eastern AB (Dan et al., 2014; Geng, Wang, Shen, & Wu, 2002; Geng & Zhou, 2010, 2011); and those determined in this paper. However, the NCC seemed to be in a within-plate environment with almost no magmatic events. Detrital zircon data also proved this, that the detrital zircons from strata of the AB are extensively developed with an ~ 900 Ma age peak, but detrital zircons from strata of the NCC did not have this characteristic.

In fact, when and how did the AB deviated from the NCC (Ordos Block) still remains unclear; the absence of basic rocks or bimodal volcanic rocks found in the eastern AB makes the evidence insufficient; we can only obtain the information that the AB is independent from the NCC during the Neoproterozoic because the ~ 900 Ma age peak

of detrital zircons are absent in the NCC. However, many researchers consider that the AB was amalgamated onto the NCC again during the Paleozoic; they put forward multiple amalgamation periods as in the Ordovician, Devonian, or Carboniferous (Dan et al., 2016; Liu, Zhao, Liu, & Shi, 2017; Yuan & Yang, 2015; Zhang et al., 2015; Zhang, Zhang, Zhang, et al., 2016; Zhang, Zhang, & Zhao, 2016), but the age of the deviation mentioned above is indistinct. In this study, we consider that the AB was deviated from the NCC after ~2.0 Ga; because they have similar detrital zircon age peak at this time, they were likely to have experienced common geological events at this time as an amalgamated plate; however, our detrital zircons indicate that the AB was in an extension-rift environment during 900–800 Ma inside the RS, separated from the NCC. Thus, we believe that the AB may deviate from the NCC between ~2.0 and ~900 Ma: a large span Proterozoic period, but most probably in the Mesoproterozoic.

Li et al. (2005) considered that the ~830 Ma mafic-ultramafic rock from the Jinchuan and Longshoushan areas had the same characteristics as the mafic-ultramafic rock in the YB during the same period: They were all in an extension-rift environment, before or during the period of the RS breakup. The AB became a part of the RS before the Neoproterozoic and may be located beside the YB. Although the detrital zircons from the northern YB also recorded abundant magmatic events during the Neoproterozoic, the zircon characteristics are different from the AB. In the diagram of the detrital zircon $\epsilon\text{Hf}(t)$ values vs. ages (Figure 15), the detrital zircons from the AB and northern YB during the Neoproterozoic present different isotopic contents: The detrital zircons from the AB have an age peak between 1,000 and 800 Ma, with $\epsilon\text{Hf}(t)$ below zero; this may be related to the ancient lithospheric mantle under the AB. The detrital zircons may have been from the 1,000–800 Ma magmatic events from the Longshoushan area, such as the zircons reported in this study; however, detrital zircons in the northern YB have an age peak between 800 and 700 Ma, with $\epsilon\text{Hf}(t)$ above zero. Some may approach the depleted mantle line, which indicates a depleted mantle or juvenile

crust origin. This is more similar to the zircons from igneous rocks in the Jiangnan Orogen, rather than the AB. Thus, we consider that the detrital zircons from the Neoproterozoic are possibly from the magmatic events that were related to the amalgamation between the YB and Cathaysia Block, which do not match the 1,000–800 Ma extension-rift related magmatic events. Although the 1,000–800 Ma magmatic events from the AB and 800–700 Ma magmatic events from the northern YB may both be related to the RS breakup, it is obvious that the location of the AB in the RS may not have been beside the YB.

Compared with the North Qilian Orogen, the detrital zircons from the AB and North Qilian Orogen both record three age peaks at ~2.5 Ga; 1,000–900 Ma, and 500–400 Ma. The 500–400 Ma age peak is the maximum in the North Qilian Orogen, which may be related to the magmatic events that resulted from the subduction of Qilian Ocean to the AB, such as the ~460 Ma arc environment syenite (Zeng, 2016). In addition, the ~2.5 Ga and 1,000–900 Ma age peak is also similar to the AB and North Qilian Block. It seems that the AB and North Qilian Orogen both experienced the same geological events from the Paleoproterozoic to the Paleozoic. Wang et al. (2009) and Ge and Liu (2000) compared the palaeontology, stratigraphy, paleogeography, and tectonic thermal events of the AB, Qilian Block, Tarim Block, and Qaidam Block and suggested that they used to be one uniform block, which they called western craton. However, Zeng (2016) and references therein believe that the Qilian Ocean did exist during the Ordovician; the Qilian Orogen was formed by the collision of the Qilian-Qaidam Block and AB, which occurred after the closure of the Qilian Ocean during the Ordovician to Silurian. Even so, the age peaks of detrital zircons from the AB and North Qilian Orogen are nearly the same, and their affinities cannot be denied. However, were they a uniform block the entire time, or had they been separated during some period? This needs further study to confirm.

In conclusion, we obtained the tectonic evolutionary history of the AB from the Paleoproterozoic to the Paleozoic as follows (Figure 16):

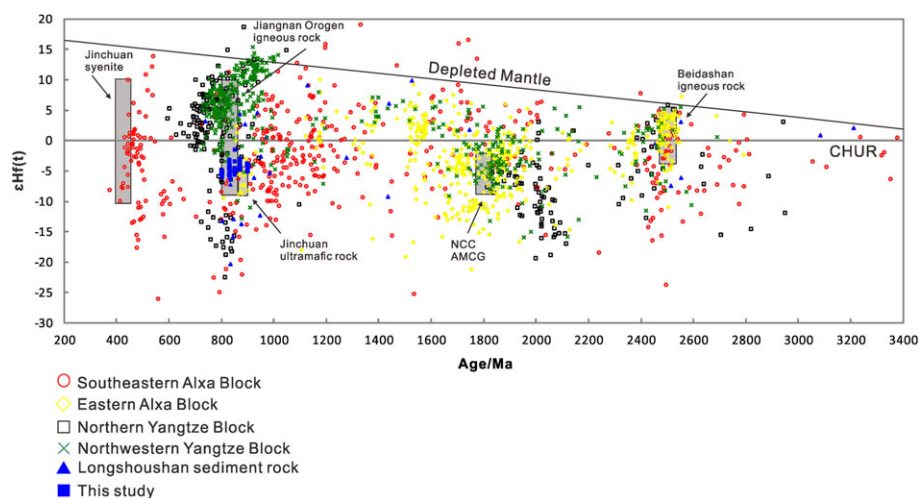


FIGURE 15 Detrital zircon $\epsilon\text{Hf}(t)$ values vs. ages diagram. Plots of detrital zircons are from the south-eastern Alxa Block (Yuan & Yang, 2015); the eastern Alxa Block (Liu et al., 2017); the northern Yangtze Block (Wang et al., 2013); the north-western Yangtze Block (Sun et al., 2009); and Longshoushan area (Tang et al., 2014). Grey area represent igneous rock zircons are from Jinchuan syenite (Zeng et al., 2016); Jiangnan Orogen igneous rock (Xin et al., 2017); Jinchuan ore-bearing ultramafic rock (Jiao et al., 2014; Jun et al., 2016); NCC AMCG (Zhang et al., 2007; Zhang, Zhao, & Santosh, 2012); and Beidashan igneous rock (Zhang, Gong, Yu, Li, & Hou, 2013) [Colour figure can be viewed at wileyonlinelibrary.com]

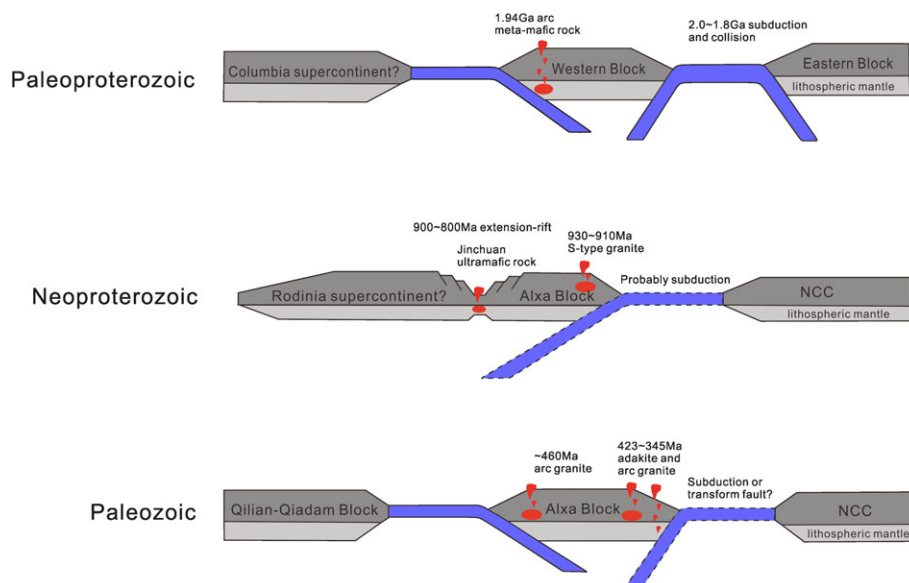


FIGURE 16 Tectonic evolution cartoon diagram of the Alxa Block [Colour figure can be viewed at wileyonlinelibrary.com]

During the Paleoproterozoic, the AB may have experienced an amalgamation onto the NCC. Previous researchers have proven that the Jinchuan mineral area, which is in the southwest of the AB, was in a subduction-collision environment, according to the ~1.9 Ga arc meta-mafic rock and 1.9–1.8 Ga migmatization (Zeng, 2016). However, many researchers also believe that the NCC were amalgamated by many microblocks at 2.0–1.8 Ga along the Central Orogenic Belt and Khondalite Series (Kusky, 2011; Peng et al., 2014; S. H. Zhao et al., 2012). Thus, we consider that the AB may have participated in the amalgamation of the NCC, which is related to the amalgamation of the Columbia supercontinent. Although the ~2.5 Ga magmatic event extensively developed in the AB and NCC is recorded, this may match a more ancient global tectonic event that we do not discuss here. However, the 1,000–900 Ma magmatic events are not recorded in the NCC, which indicates that the AB may have separated from the NCC during the Mesoproterozoic.

During the Neoproterozoic, the AB was a part of the RS (Li et al., 2005; Zeng, 2014, 2016). The Jinchuan mineral area and Longshoushan area, which are southwest of the AB, experienced a massive amount of 900–800 Ma extension-rift magmatic events; these match the global magmatic events related to the breakup of the RS. Moreover, the discovery of the 930–910 Ma S-type granitic gneiss in the eastern AB indicates a compressional environment (Dan et al., 2014), which probably matches a continental convergence and oceanic crustal subduction.

During the Paleozoic, a series of 500–400 Ma arc to post-collision igneous rocks in the south-western AB and ophiolite suite in the North Qilian Orogen proved the closure of the Qilian Ocean and collision between the Qilian-Qaidam Block and AB (Zeng, 2016; Zeng et al., 2016). However, the amalgamation of the AB and NCC may not have been accomplished until the Neopaleozoic (Zhang, Zhang, Zhang, et al., 2016; Zhang, Zhang, & Zhao, 2016). Although the ophiolite suite was not found, a series of 423–325 Ma high Sr/Y adakite and arc granites were discovered in the eastern AB (Dan et al., 2016). Some authors also believe that the amalgamation already

occurred during the Ordovician because of the shallow-water sediment rocks deposited between the AB and NCC (Xu et al., 2015). Considering the evidences of detrital zircons, sedimentary strata, and igneous rocks, all of these discoveries may provide evidences that the ocean closure and continental amalgamation occurred between the AB and NCC, although the precise time is controversial.

In fact, some authors consider that the AB had an affinity with the Gondwana land, similar to many microblocks, but different with the NCC. The AB was deviated from the original Gondwana land inside the RS at 1,000–900 Ma; the rest of the RS was broken up into several lands including the original Gondwana land, while the NCC was far away from them (Li, Zhao, Li, et al., 2016; Li, Zhao, Yu, et al., 2016; Li et al., 2017). During the Ordovician, the NCC, the AB, and many microblocks moved parallel and docked with each other by transform faults (Figure 16), constituted an uniform block called the Asian Hunic Superterrane, they were separated with the Gondwana land by the Proto-Tethys Ocean and gradually amalgamated onto the Gondwana land during 475–460 Ma (Li et al., 2017; Mu et al., 2018; Xu et al., 2015). Until 380 Ma, the Gondwana land in the south and the Laurussia land in the north were gradually amalgamated to constitute the Pangea supercontinent (Li et al., 2017).

7 | CONCLUSIONS

1. Two types of metamorphic rock were found in the Jinchuan mineral area, which are oxidated ultramafic rock and chlorite tremolite schist. Multiple zircons with different ages were found in the oxidated ultramafic rock, which were mostly inherited zircons with ages greater than 1,000 Ma and hydrothermal zircons with ages below 800 Ma. Multiple zircons with different ages were also found in chlorite tremolite schist, except the inherited zircons. Most zircons were magmatic zircons with ages between 900 and 800 Ma, and several hydrothermal zircons with ages below 800 Ma were also present.

2. The 900–800 Ma magmatic zircons from the chlorite tremolite schist exhibited different morphologies, which indicates multiple sources. Their host rocks were mainly larvikite and dolerite, which were formed in a within-plate environment. The magma may originate from the remelting of enriched lithospheric mantle and experienced a significant degree of crystallization differentiation. The 900–800 Ma within-plate host rocks recorded at least a 100 Myr extension-rift environment in the AB.
3. According to the comparison of detrital zircons, we consider that the AB may have amalgamated onto the NCC and the Columbia supercontinent since the Paleoproterozoic. The AB was part of the RS, but not located beside the YB. The AB started to separate from the RS 900–800 Ma, but the whole breakup may have occurred later than 800 Ma. The closure of the Qilian Ocean and collision between the AB and Qilian-Qaidam Block occurred during the Ordovician, while the AB was amalgamated onto the NCC during the Neoproterozoic.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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